

Recommendation Supervision Semantics and Exposure-Aware Evaluation for Recommendation-Tuned LLMs

推荐监督语义与面向曝光量的推荐调优 LLM 评估

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Abstract—Empirical conclusions about large language models (LLMs) for recommendation can depend on what is supervised, how much downstream supervision a model consumes, and how candidates are evaluated. We study these dependencies using Llama-3.2-3B-Instruct with QLoRA, taking MovieLens-1M as the main dataset and Amazon Musical Instruments as a directional ranking-side check. We compare rating-derived preference prediction (Y), next-interaction prediction (N), and a shared Y+N multitask adapter (M1), trace task-sample exposure, evaluate alternative candidate protocols, and compare N-trained LlamaRec with SASRec at approximately matched downstream exposures. The complete Y and N formulations yield distinct measured capability profiles. In the MovieLens seed42 trajectory, Y’s measured gains through 96k are limited or uneven, whereas N ranking improves across measured validation and test points through 200k. On a common cross-task-safe subset, the standard-k5 N-specialist-M1 validation gap narrows from 48k to 96k per task across all three metrics, although test does not reproduce the narrowing. At 96k, three training seeds retain a small but consistently positive N advantage, while M1 preserves comparable Y-side preference capability. Alternative candidate protocols materially change the remaining specialist-multitask gap. N-trained LlamaRec leads the evaluated SASRec baseline at four approximately matched downstream task-sample exposure points, but SASRec improves strongly and the gap narrows at 200k. Safe ranking directions also appear on Amazon at earlier seed42 operating points. These results show that recommendation capability and relative model performance are conditional on prediction formulation, cumulative downstream exposure, and candidate evaluation protocol.

关于大语言模型推荐的经验结论，可能取决于模型接受何种监督、消耗多少下游监督，以及如何评估候选物品。本文采用 Llama-3.2-3B-Instruct 与 QLoRA，以 MovieLens-1M 为主要数据集，并以 Amazon Musical Instruments 作为排序侧方向性检验。研究比较评分产生的偏好预测 Y、下一交互预测 N 和共享 Y+N 多任务 adapter M1，追踪任务样本曝光，考察不同候选协议，并在下游曝光近似匹配时比较接受 N 任务训练的 LlamaRec 与 SASRec。完整的 Y 与 N 预测形式呈现不同的被测能力结构。在 MovieLens seed42 轨迹中，Y 到 96k 的被测增益有限或不均匀，N 排序则在验证集和测试集的已测点持续改善至 200k。在共同 cross-task-safe 子集上，标准 k5 的 N 专家-M1 验证差距在三个指标上均从每任务 48k 到 96k 收窄，但测试集没有复现这一收窄。96k 时，三个训练 seed 均保留小幅且方向一致的 N 优势，M1 同时保持与 Y 专家接近的被测偏好能力。其他候选协议明显改变了剩余的专家与多任务差距。接受 N 任务训练的 LlamaRec 在四个下游任务样本曝光近似匹配点均领先被评估的 SASRec 基线，但 SASRec 随监督增加明显改善，200k 时差距缩小。Amazon 的较早 seed42 运行点也呈现满足安全条件的排序方向。这些结果表明，对推荐能力和模型相对表现的解释应结合预测形式、累计下游曝光和候选评估协议。

Index Terms—Large language model recommendation, Recommendation supervision, Sequential recommendation, Multitask learning, Task-sample exposure

I. Introduction / 引言

Language-model-based recommendation spans zero-shot conversation, candidate ranking, and recommendation-specific adaptation [1]–[3]. Recommendation-as-language-processing further offers a shared language interface for model inputs and predictions [4]. Because language models can interact, rank, or adapt to a task, interpreting what recommendation-specific supervision teaches them is a central empirical problem [5].

基于语言模型的推荐涵盖零样本对话、候选排序和推荐特定适配[1]–[3]。将推荐表述为语言处理，也为模型输入与预测提供了共同语言接口[4]。语言模型可以承担交互、排序或任务适配等职责，因此，推荐特定监督究竟教会模型何种能力，是一个核心实证问题[5]。

Recommendation includes different prediction questions. Whether a user likes an item is a preference judgment; which item appears next is a subsequent-event prediction. Feedback and sequential models operationalize different targets, while language frameworks can express both through a common interface [4], [6], [7]. For an adapted model, task data, prompt, target, and scoring interface jointly define the complete supervision formulation and the capability observed.

推荐包含不同的预测问题。用户是否喜欢某个物品属于偏好判断，下一次出现哪个物品则属于后续事件预测。反馈模型与序列模型将不同目标具体化，语言框架可以用共同接口表达两者[4], [6], [7]。对于适配模型，任务数据、提示、目标与评分接口共同构成完整监督形式，并决定被观察的能力。

A checkpoint also represents a specific amount of adaptation. We define task-sample exposure as the cumulative number of examples consumed for a task, including repetitions. Unified recommendation studies already examine task performance and changes in data amount or composition [4], [8]. Our complementary question is how specialist-shared relationships evolve as the corresponding task receives more cumulative supervision, with total and per-task exposure separated for mixed-task training.

一个检查点也代表特定程度的适配。本文将任务样本曝光定义为某任务累计消耗的样本数，包括重复样本。统一推荐研究已考察任务表现以及数据量或构成的变化[4], [8]。本文进一步研究对应任务获得更多累计监督时，专家

与共享模型的关系如何演变，并在混合任务训练中区分总曝光和每任务曝光。

Evaluation conditions add another boundary. Sampled metrics, sampling strategies, and candidate construction can change measured performance and model orderings [9]–[11]. Task formulation and data allocation likewise intersect in unified recommendation and shared search-recommendation models [4], [8], [12], [13]. These precedents motivate jointly examining cumulative per-task supervision, specialist-shared modeling, and candidate protocol without treating any one as an isolated cause.

评估条件构成另一项边界。采样指标、采样策略和候选构造能够改变测得的表现及模型次序[9]–[11]。在统一推荐和搜索推荐共享模型中，任务形式与数据分配也相互关联 [4], [8], [12], [13]。这些先例促使本文联合考察累计每任务监督、专家与共享建模及候选协议，而不把其中任何一项视为孤立原因。

We conduct a controlled study with rating-derived preference supervision (Y), next-interaction supervision (N), and one shared Y+N adapter (M1). We trace task-specific exposure, compare specialists with M1 at matched per-task exposure, test alternative candidate protocols, and compare N-trained LlamaRec with SASRec at approximately matched downstream supervision. MovieLens supplies detailed trajectories; Amazon Musical Instruments provides an external ranking-side check.

本文开展一项受控研究，比较评分产生的偏好监督 Y、下一交互监督 N 和共享 Y+N adapter M1。研究追踪任务特定曝光，在每任务曝光对齐时比较专家与 M1，检验不同候选协议，并在下游监督近似匹配时比较 N 训练的 LlamaRec 与 SASRec。MovieLens 提供详细轨迹，Amazon Musical Instruments 提供外部排序侧检验。

The complete Y and N formulations produce distinct measured capability profiles. In the seed42 MovieLens trajectory, Y gains are limited or uneven, whereas N improves through the evaluated range. On a common cross-task-safe subset, the standard-k5 validation specialist-M1 gap narrows from 48k to 96k per task across all three metrics; two additional seeds reproduce a small positive N advantage at 96k, but the common seed42 test subset does not reproduce the narrowing. Alternative protocols retain an N advantage with protocol-dependent magnitude. Ntrained LlamaRec leads SASRec at four approximately matched exposure points, while SASRec improves and the gap narrows at 200k. Amazon supports safe earlier-point ranking directions only.

完整 Y 与 N 形式呈现不同的被测能力结构。在 MovieLens 的 seed42 轨迹中，Y 的增益有限或不均匀，N 则在已评估范围内持续改善。在共同 cross-task-safe 子集上，标准 k5 验证中的专家-M1 差距在三个指标上均从每任务 48k 到 96k 收窄；另外两个 seed 复现了 96k 时小幅的 N 优势，但共同 seed42 测试子集没有复现收窄。其他候选协议也保持 N 方向，幅度随协议变化。N 训练的 LlamaRec 在四个曝光近似匹配点领先 SASRec，同时 SASRec 持续改善，差距在 200k 缩小。Amazon 仅支持满足安全条件的较早运行点排序方向。

Our primary contribution is an exposure-conditioned account of specialist-shared modeling. The secondary contribution is a four-point cumulative downstream-exposure comparison between the LLM and a sequential baseline. Candidate-protocol analysis supplies a supporting boundary, while supervision-formulation distinctions frame the capabilities being compared. Together, they show that conclusions depend on what is supervised, how much task supervision is consumed, and how candidates are evaluated.

本文的主贡献是结合曝光刻画专家与共享建模关系；次贡献是在四个累计下游曝光点比较 LLM 与序列基线；候选协议分析提供支撑性边界，监督形式区分则界定被比较的能力。三者共同表明，结论取决于监督内容、累计任务监督量和候选评估方式。

II. Related Work / 相关工作

A. LLMs for Recommendation / 面向推荐的语言模型

Prompt-based recommendation places language models directly in recommendation requests. Zero-shot conversational recommendation and zero-shot ranking emphasize different outputs: interaction and item ordering [1], [2]. Within the broader LLM-recommender literature [5], this diversity makes the requested output part of what an evaluation measures and motivates task-specific capability claims.

基于提示的推荐使语言模型直接处理推荐请求。零样本对话推荐与零样本排序强调不同输出，即交互和物品排序 [1], [2]。在更广泛的 LLM 推荐研究中[5]，这种差异使所要求的输出成为评估对象的一部分，也要求按任务陈述能力。

Recommendation-specific adaptation makes the training interface part of the model’s role. P5 expresses multiple task families as language pairs, TALLRec trains Yes/No recommendation responses, and InstructRec supports pointwise, pairwise, matching, and reranking requests [3], [4], [14]. These precedents establish language-based adaptation across tasks. We compare the capabilities measured after complete preference and next-interaction formulations under a shared model setting.

推荐特定适配使训练接口成为模型职责的一部分。P5 将多个任务族表述为语言输入输出对，TALLRec 训练 Yes/No 推荐响应，InstructRec 支持逐点、成对、匹配和重排序请求[3], [4], [14]。这些先例已覆盖多任务语言化适配；本文则在共同模型设置下比较完整偏好与下一交互形式适配后的被测能力。

B. Recommendation Supervision and Task Formulation / 推荐监督与任务形式

Recommendation targets differ before any language interface. Matrix factorization fits ratings, implicit-feedback methods distinguish preference from confidence in observations, and sequential models predict a subsequent item [6], [7], [15]. Their scores therefore concern assessments, observed behavior, or next-event selection. Separating target from modeling technique clarifies which capability a score supports.

在引入语言接口之前，推荐目标就有区别。矩阵分解拟合评分，隐式反馈方法区分偏好与行为观测置信度，序列模型则预测后续物品[6], [7], [15]。相应分数分别涉及评价、观测行为或下一事件选择。区分目标与建模技术，才能明确分数支持何种能力判断。

A common language interface can retain these distinctions. P5 includes rating-threshold like/dislike questions and next-item interfaces; ITDR also combines rating and next-item tasks [4], [8]. Task distinction is therefore not new. Our framing compares their observed native and bridge capability profiles while treating target, task data, prompt, and scoring rule as one complete formulation rather than a single isolated semantic factor.

共同语言接口仍可保留这些区别。P5 同时包含评分阈值下的喜欢与否问题及下一物品接口，ITDR 也组合评分与下一物品任务[4], [8]。任务区分本身并非本文首次提出；本文比较其原生与桥接能力结构，并将目标、任务数据、提示和评分规则视为完整形式，而非单一语义因素。

C. Multitask and Unified Recommendation Modeling / 多任务与统一推荐建模

Unified recommendation already links task design, shared modeling, and data allocation. P5 varies derived instances while comparing single- and multitask models; ITDR reports task-removal and data-subset studies; OpenOneRec specifies task-sampling weights and contrasts low-data, single-domain, and multidomain adaptation [4], [8], [12]. These are substantive data-aware multi-task precedents. Our narrower question tracks specialist-shared relationships as corresponding-task cumulative exposure changes.

统一推荐已将任务设计、共享建模和数据分配联系起来。P5 在比较单任务与多任务模型时改变衍生实例数，ITDR 报告任务删除与数据子集实验，OpenOneRec 明确任务采样权重并比较低数据、单域与多域适配[4], [8], [12]。这些都是数据感知多任务研究的实质先例；本文更具体地追踪对应任务累计曝光变化时的专家-共享关系。

Specialization also depends on the combined tasks or domains. Penha et al. vary added-task per-item caps when comparing specialized and shared search-recommendation models, while OneReason studies reinforcement-learning experts and a unified student [13], [16]. Rather than claiming the first data-aware specialist-shared comparison, we align cumulative exposure for each Y/N target task and then test how the resulting ranking relationship varies across candidate protocols.

专门化还取决于被组合的任务或领域。Penha 等人在比较搜索推荐专家与共享模型时改变新增任务的单物品实例上限，OneReason 则研究强化学习专家与统一学生[13], [16]。本文不声称首次进行数据感知的专家-共享比较，而是对齐 Y/N 对应目标任务的累计曝光，再检验所得排序关系如何随候选协议变化。

D. Evaluation and Baseline Comparison / 评估与基线比较

Candidate construction can alter offline ranking conclusions: sampled metrics need not preserve exact-metric orderings, and sampling strategy can affect measurements and

model rankings [9], [11]. Target-set construction and difficult retrieved candidates provide related precedents [10], [14]. Candidate sensitivity is therefore established. We use separate non-nested protocols to delimit the higher-exposure specialist-multitask relationship, without isolating candidate count from composition or sampling.

候选构造能够改变离线排序结论：采样指标不必保留精确指标的模型次序，采样策略也会影响测量结果和模型排序[9], [11]。target set 构造与困难检索候选已有相关先例 [10], [14]。因此候选敏感性并非新发现；本文以独立非嵌套协议限定较高曝光下的专家-多任务关系，不把候选数量与组成或采样方式隔离开来。

Training conditions complement evaluation scope. SASRec is a sequential reference, while replicability and standardized-evaluation studies underscore careful tuning and the competitiveness of conventional baselines [7], [17], [18]. TALLRec already compares an adapted LLM with SASRec under selected-example budgets, while sequencescaling work studies data-pool size and repetition [3], [19]. Our contribution is not the first budget-aware comparison: it follows both models over four approximately matched cumulative downstream-consumption points, a quantity distinct from selected examples and computation.

训练条件补充了评估范围。SASRec 提供序列参照，复现性与标准化评价研究强调谨慎调优以及传统基线的竞争力 [7], [17], [18]。TALLRec 已按选出样本预算比较适配 LLM 与 SASRec，序列缩放研究也考察数据池规模和重复训练 [3], [19]。本文并非首次开展预算感知比较，其贡献是在四个累计下游消费近似匹配点追踪两类模型；这一口径不同于选出样本数和计算量。

Prior work therefore establishes task diversity, shared modeling, data-dependent performance, and candidate-sensitive evaluation. We connect these themes by tracing specialist-shared relationships over cumulative per-task supervision, using a sequential-baseline trajectory and alternative candidate protocols as complementary comparisons and boundaries. Supervision formulation specifies the prediction capability to which each conclusion applies.

已有研究已经建立任务多样性、共享建模、数据依赖表现和候选敏感评估等主题。本文通过累计每任务监督下的专家-共享关系将其连接起来，并以序列基线轨迹和不同候选协议提供互补比较与边界。监督形式则明确每项结论所对应的预测能力。

III. Problem Formulation / 问题定义

Let an interaction be a user-item event with a rating and timestamp, and let $H(u, t)$ contain a user's interactions strictly before t . Samples use the complete sequence, while model input retains at most the 10 most recent strictly earlier interactions, $H_{10}(u, t)$, paired with an item i or candidate set C . A rating-derived preference label and the next observed item answer different questions even when drawn from the same sequence.

一次交互由用户、物品、评分和时间戳构成， $H(u, t)$ 表示用户在 t 之前发生的交互。样本构造使用完整序列，模型输入最多保留严格早于目标时刻的最近 10 条交互

$H_{10}(u,t)$, 并与物品 i 或候选集 C 配对。即使来自同一序列, 评分产生的偏好标签与下一观测物品回答的也是不同问题。

A. Preference, Next Interaction, and Multitask Interfaces / 偏好、下一交互与多任务接口

Y estimates $P(\text{Like} | H_{10}, i)$, where ratings at least 4 map to Yes and lower ratings to No. The complete sequence, including low-rated events, supplies histories and targets. Native evaluation measures rating-derived class discrimination; independently scoring candidates through $P(\text{Yes})$ provides the bridge to next-interaction ranking.

Y 估计 $P(\text{Like} | H_{10}, i)$, 评分不低于 4 映射为 Yes, 较低评分映射为 No。包含低评分事件的完整序列用于构造历史与目标。原生评估衡量评分标签判别, 通过 $P(\text{Yes})$ 独立评价候选则形成通向下一交互排序的桥梁。

N estimates $P(\text{NextInteraction} = i | H_{10}, C)$, where C contains the observed next item and fixed distractors; the answer is the target's candidate label. Targets use all rating levels. A distractor is an alternative to the observed next event, not a confirmed dislike, and previously encountered items remain eligible.

N 估计 $P(\text{NextInteraction} = i | H_{10}, C)$, 其中 C 包含观测下一物品和固定干扰候选, 答案为目标的候选标签。目标覆盖全部评分等级。干扰候选是观测下一事件的备选项, 而非确认的不喜欢; 用户曾接触的物品仍可作为候选。

$M1$ interleaves Y and N within one shared adapter while retaining task-specific prompts and answer spaces. $M-Y$ and $M-N$ denote its two scoring interfaces. Each is compared with the corresponding specialist at matched task exposure; $M1$ total exposure is the sum across tasks.

$M1$ 在一个共享 adapter 中交错使用 Y 与 N , 同时保留任务特定提示和答案空间。 $M-Y$ 与 $M-N$ 表示两条评分接口, 各自与对应任务曝光匹配的专家比较; $M1$ 总曝光为两项任务曝光之和。

$RQ1$ compares the observed Y and N capability profiles; $RQ2$ traces their response to task-sample exposure. $RQ3$ asks how closely $M1$ approaches both specialists at matched per-task exposure, and $RQ4$ tests how the standard PopMatch- $k5$ relationship changes under alternative candidate protocols. $RQ5$ follows the exposure-conditioned comparison between N -trained LlamaRec and SASRec.

$RQ1$ 比较 Y 与 N 的可观测能力结构, $RQ2$ 追踪其对任务样本曝光的响应。 $RQ3$ 考察每任务曝光对齐时 $M1$ 与两种专家的接近程度, $RQ4$ 考察标准 PopMatch- $k5$ 关系在其他候选协议下如何变化。 $RQ5$ 追踪曝光条件下 N 训练的 LlamaRec 与 SASRec 之间的比较。

IV. Methodology / 研究方法

The pipeline constructs temporally legal examples, adapts a common base model with task-specific prompts, and scores allowed responses. Complete user sequences define examples, bounded recent histories form prompts, and the Y and N interfaces produce preference scores or candidate orderings.

实验流程构造时间合法样本, 以任务特定提示适配共同底座, 并为允许响应评分。完整用户序列用于定义样本, 受限近期历史构成提示, Y 与 N 接口分别输出偏好分数或候选排序。

A. Temporal Sample Construction / 时间约束下的样本构造

Each user's sequence is grouped into ordered timestamp buckets. A target at t uses only events strictly before t , so same-bucket Y targets share the preceding history. The last bucket supplies Y test targets, the previous bucket supplies validation, and earlier buckets supply training; a held-out bucket may yield several Y targets. Prompts retain the 10 most recent eligible events.

每位用户的序列按时间戳分为有序时间桶。时刻 t 的目标只使用严格早于 t 的事件, 因此同桶 Y 目标共享此前历史。末桶提供 Y 测试目标, 前一桶提供验证目标, 更早时间桶提供训练样本; 一个留出桶可产生多个 Y 目标。提示保留最近 10 条合法事件。

N requires preceding history and one unique next item, so legal targets are single-interaction buckets after the initial bucket. Multi-interaction buckets are skipped as targets but remain in later histories. The last two legal N examples become validation and test; earlier legal examples form training. Inputs again retain at most 10 strictly earlier interactions.

N 同时要求先前历史和唯一下一物品, 因此合法目标来自首桶之后的单交互时间桶。多交互桶不作为目标, 但其事件仍进入后续历史。末两个合法 N 样本依次作为验证和测试, 更早合法样本用于训练; 输入同样最多保留 10 条严格更早交互。

B. Shared Adaptation and Response Scoring / 共享适配流程与响应评分

We adapt Llama-3.2-3B-Instruct [20] with QLoRA over a frozen 4-bit NF4 base [21]. Rank 16 LoRA adapters [22] use alpha 32 and dropout 0.05 across $q/k/v/o$ and gate/up/down projections. Gradient checkpointing is enabled and maximum sequence length is 2,048.

本文以 QLoRA 适配 Llama-3.2-3B-Instruct [20], 底座采用冻结 4-bit NF4 表示[21]。rank 16 的 LoRA adapter [22] 使用 alpha 32、dropout 0.05, 覆盖 $q/k/v/o$ 及 gate/up/down 投影。训练启用 gradient checkpointing, 最大序列长度为 2,048。

Examples use instruction-chat format with the answer as the assistant response. Prompt and padding tokens are masked, giving response-only loss. Y supervises Yes/No and N supervises the target candidate label. $M1$ shares adapter parameters and interleaves Y/N examples 1:1, with the prompt identifying the task.

样本采用指令对话格式, 答案作为 assistant 响应。提示与 padding token 被屏蔽, 仅响应部分参与损失。 Y 监督 Yes/No, N 监督目标候选标签。 $M1$ 共享 adapter 参数, 以 1:1 交错 Y/N 样本, 并由提示标识任务。

Inference scores allowed-answer likelihoods. Y normalizes Yes/No likelihoods and Y -as-ranker orders independently

scored candidates. N normalizes candidate-label likelihoods conditioned on the list. Multi-token answers use complete-sequence log-likelihood; M-Y and M-N reuse these routes.

推理对允许答案的似然评分。Y 归一化 Yes/No 似然，Y-as-ranker 据独立候选分数排序；N 在候选列表条件下归一化候选标签似然。多 token 答案使用完整序列对数似然，M-Y 与 M-N 复用相应路径。

One-device micro-batch 1 with 8 accumulation steps yields 8 examples per optimizer update. Checkpoints are indexed by cumulative steps and task exposure and evaluated on fixed candidates. Validation guides training decisions; test follows frozen selection. Multitask per-task allocation is specified below.

TABLE I Task-specific dataset splits. / 按任务分别统计的数据划分。

Dataset	Task	Users	Train	Validation	Test
MovieLens-1M	Y	6040	976284	12381	11544
MovieLens-1M	N	5675	212725	5675	5675
Amazon Musical Instruments	Y	57439	396908	57442	57442
Amazon Musical Instruments	N	57439	339449	57439	57439

MovieLens-1M 用于曝光及专家-多任务分析，Amazon Musical Instruments 提供外部排序检验。原生偏好与排序指标分开报告，并标明划分和曝光。验证集指导曝光与检查点决策，冻结测试结果仅在这些决策完成后报告。

A. Datasets and Evidence Coverage / 数据集与证据覆盖

MovieLens-1M [23] uses task-specific temporal splits summarized in Table I. Different target and timestampbucket rules yield 6,040 Y users and 5,675 legal N users. Ratings cover 3,706 distinct items, while the metadata universe contains 3,883, a distinction retained for coverage interpretation.

MovieLens-1M [23] 采用 Table I 汇总的任务特定时间划分。不同目标与时间桶规则产生 6,040 名 Y 用户和 5,675 名合法 N 用户。评分覆盖 3,706 个不同物品，元数据全集包含 3,883 个物品，这一区别用于解释覆盖范围。

The processed Amazon Reviews 2023 5-core Musical Instruments data [24] contain 57,439 users, 24,584 items, and

单设备 micro-batch 为 1、梯度累积为 8，每次优化器更新消耗 8 个样本。检查点按累计步数和任务曝光标识，并在固定候选上评估。验证集指导训练决策，测试在选择冻结后进行；多任务每任务分配见下文。

V. Experimental Setup / 实验设置

MovieLens-1M supports the exposure and specialist-multitask analyses; Amazon Musical Instruments supplies an external ranking check. Native preference and ranking metrics are reported separately with split and exposure identified. Validation guides exposure and checkpoint decisions, while frozen-test results are reported only after those decisions.

511,792 interactions; task splits appear in the same table. External evidence uses seed42 full-test Random-k5 and PopMatch-k5 ranking at available earlier operating points, covering Y-as-ranker, N, and sequential references.

处理后的 Amazon Reviews 2023 5-core Musical Instruments 数据[24] 包含 57,439 名用户、24,584 个物品和 511,792 次交互，任务划分见同一表。外部证据采用现有较早运行点的 seed42 完整测试集 Random-k5 与 PopMatchk5 排序，覆盖 Y-as-ranker、N 及序列参照。

B. Task-Sample Exposure and Matching / 任务样本曝光与对齐

Task-sample exposure counts examples consumed for task q, including repetitions: $E_q = \sum_s \sum_{(x \text{ in } B_s)} 1[\text{task}(x)=q]$, where B_s is optimizer step s's batch. It measures received supervision, not unique interactions, because examples may repeat and task histories overlap.

TABLE II LLM optimizer steps and task-sample exposure (effective batch 8). For a specialist, total exposure equals its task exposure. M1 total exposure is twice its expected per-task exposure, conditional on correct resume data skipping. / LLM 优化器步数与任务样本曝光（有效 batch 为 8）。专家模型的总曝光等于对应任务曝光；M1 总曝光为预期每任务曝光的两倍，该统计以续训正确跳过已读数据为条件。

Run	Steps	Total	Y exposure	N exposure
Y24	3000	24000	24000	0
Y48	6000	48000	48000	0
Y96	12000	96000	96000	0
N24	3000	24000	0	24000
N48	6000	48000	0	48000
N96	12000	96000	0	96000
N200	25000	200000	0	200000
M1-48	12000	96000	48000	48000
M1-96	24000	192000	96000	96000

任务样本曝光统计任务 q 累计消耗的样本，包括重复样本： $E_q = \sum_s \sum_{(x \text{ in } B_s)} 1[\text{task}(x)=q]$ ，其中 B_s 为第 s 次优化器更新的 batch。由于样本可重复且任务历史重叠，它衡量接受的监督量，而非唯一交互数。

With effective batch 8, specialist updates map directly to task exposure. Balanced M1-48 and M1-96 receive expected per-task exposures of 48k and 96k and totals of 96k and 192k; this accounting assumes correct resume data skipping. Table II separates optimizer steps, total exposure, and Y/N exposure.

有效 batch 为 8 时，专家模型更新数直接对应任务曝光。均衡 M1-48 与 M1-96 的预期每任务曝光分别为 48k 和 96k，总曝光为 96k 和 192k；该统计以续训正确跳过已读数据为前提。Table II 分别列出优化器步数、总曝光和 Y/N 曝光。

N200 consumes 200,000 examples, about 94.0% of the 212,725-example legal N training pool, and serves as a near-one-pass anchor. SASRec uses recorded actual consumption, including short final batches: 24,064, 48,128, 96,256, and 200,000 examples. These points approximately match N24/N48/N96/N200 rather than equate computation.

N200 消耗 200,000 个样本，约占 212,725 个合法 N 训练样本的 94.0%，作为接近单遍训练的参照。SASRec 采用包含末尾不足完整 batch 在内的实际消费量：24,064、48,128、96,256 和 200,000 个样本。这些点近似匹配 N24/N48/N96/N200，并不对齐计算量。

C. Native Metrics and Candidate Protocols / 原生指标与候选协议

Y-native evaluation reports AUC, F1, and accuracy; paired 96k F1/accuracy use threshold 0.5. Y-as-ranker independently scores candidates by P(Yes), whereas Nnative and M-N use candidate-label likelihoods. Ranking reports HR@1, NDCG@5, and MRR; NDCG is truncated at 5 and MRR uses the complete candidate ranking.

Y 原生评估报告 AUC、F1 和 Accuracy，96k 配对分析的 F1/Accuracy 阈值为 0.5。Y-as-ranker 按 P(Yes) 独立评价候选，N-native 与 M-N 使用候选标签似然。排序报告 HR@1、NDCG@5 与 MRR；NDCG 截断到 5，MRR 使用完整候选排序。

Random-k5 has one target and four random distractors. PopMatch-k5 draws distractors near the target’s full-corpus popularity from a pool up to 10 times the needed size, excluding the target. Fixed ordered lists are shared within comparisons. This retrospective control reduces one popularity mismatch; it is not online retrieval.

Random-k5 包含一个目标和四个随机干扰候选。PopMatch-k5 依据全语料流行度，从至多为所需数量 10 倍的近邻池抽取干扰候选并排除目标。比较内部共享固定有序列表。这一回顾性控制减弱一项流行度失配，但不模拟在线检索。

Hard-k20 and hard-k50 evaluate N96 and M1-96 on shared examples and ordered candidates within each protocol. Their candidate sets are separately constructed and non-nested, so count and composition change together. They test protocol-conditioned robustness rather than isolate candidate size.

Hard-k20 与 hard-k50 在各协议内部以共享样本和有序候选评估 N96 与 M1-96。候选集分别构造且非嵌套，因此数量与组成共同变化。它们检验协议条件下的稳健性，而非隔离候选规模。

To ensure cross-task holdout integrity for the shared adapter, M1-N comparisons are restricted to a cross-tasksafe subset of N evaluation examples. For an example $e=(u,t)$, we retain it only when its target was not consumed as an M1 Y-training target, did not enter the history of any later M1 Y-training example, and no M1 Y-training target for user u has a timestamp at or after t . The M1 N-training branch has 0 target or history overlap with the retained N holdouts. N and M1-N are evaluated on exactly the same retained examples. Training follows the task-specific streams; this additional criterion is an evaluation-side restriction, not a joint temporal cutoff used during training.

为保证共享 adapter 的跨任务留出完整性，M1-N 比较仅在 N 评估样本的 cross-task-safe subset 上进行。对于样本 $e=(u,t)$ ，仅当其目标未被 M1 的 Y 训练分支作为训练目标消费、未进入任何更晚 M1 Y 训练样本的历史，且用户 u 不存在时间戳不早于 t 的 M1 Y 训练目标时，才予以保留。M1 的 N 训练分支与保留 N 留出集的目标及历史重叠均为 0。N 与 M1-N 在完全相同的保留样本上评估。训练仍采用各任务自己的训练流；这一附加准则只作用于评估侧，并非训练时采用的联合时间截断。

TABLE III Cross-task-safe subset coverage for M1-N comparisons on MovieLens. The restriction is applied to evaluation, and N and M1-N use the same retained examples within each row. / MovieLens 上 M1-N 比较的 cross-task-safe 子集覆盖率。该限制作用于评估侧，每行中的 N 与 M1-N 使用相同保留样本。

M1 point	Split	Retained	Full holdout	Coverage (%)
M1-48	validation	5494	5675	96.81
M1-48	test	5600	5675	98.68
M1-96	validation	5318	5675	93.71
M1-96	test	5535	5675	97.53

Cross-task-safe coverage is 5,494/5,675 (96.81%) on M1-48 validation, 5,600/5,675 (98.68%) on its test, 5,318/5,675 (93.71%) on M1-96 validation, and 5,535/5,675 (97.53%) on its test Table III. For the seed42 comparison from 48k to 96k, we use the intersection of the M1-48 and M1-96 masks, giving the same 5,318 validation and 5,535 test examples at both exposure points.

cross-task-safe subset 在 M1-48 验证集上保留 5,494/5,675 (96.81%)，测试集保留 5,600/5,675 (98.68%)；在 M1-96 验证集上保留 5,318/5,675 (93.71%)，测试集保留 5,535/5,675 (97.53%) Table III。seed42 的 48k 至 96k 比较取 M1-48 与 M1-96 mask 的交集，

使两个曝光点在相同的 5,318 个验证样本和 5,535 个测试样本上比较。

D. Sequential Baseline and Comparison Scope / 序列基线 与 比较范围

SASRec [7] uses 64-dimensional item and position embeddings, 2 attention heads, 2 causal Transformer layers, a 256-dimensional feed-forward block with GELU, dropout 0.2, final LayerNorm, and a maximum history of 10. It scores an item by the dot product between the sequence representation and item embedding plus an item bias. Training uses full-item cross-entropy and AdamW with learning rate 0.001, weight decay 0, no scheduler, batch size 512, and seed42.

SASRec [7] 采用 64 维物品与位置嵌入、2 个注意力头、2 层因果 Transformer、宽度为 256 且使用 GELU 的前馈层、0.2 dropout、末端 LayerNorm 和最长 10 条历史。物品得分为序列表示与物品嵌入的点积加物品偏置。训练采用全物品交叉熵与 AdamW，学习率 0.001、weight decay 为 0、不使用 scheduler，batch size 为 512，训练 seed 为 42。

S47, S94, S188, and S391 are predefined 47-, 94-, 188, and 391-optimizer-step operating points, independently trained from scratch rather than selected as best validation checkpoints or by early stopping. Their recorded downstream task-sample exposures are 24,064, 48,128, 96,256, and 200,000, including the short final batch. They are paired with N24/N48/N96/N200 on the same seed42 PopMatch-k5 validation and test candidate sets. Validation is the main comparison and frozen test is report-only. This design aligns downstream task-sample exposure only, not pretraining, tokens, optimizer steps, FLOPs, wall-clock time, hardware, latency, parameter count, or computational resources.

S47、S94、S188 与 S391 是预先指定的 47、94、188 和 391 个优化器步运行点，每个运行均独立从头训练，不是按验证集选择的最佳检查点，也不是早停结果。其记录的下游任务样本曝光分别为 24,064、48,128、96,256 和 200,000，包含末尾不足完整 batch 的实际消费。它们分别与 N24/N48/N96/N200 配对，并使用相同的 seed42 PopMatch-k5 验证和测试候选集。验证集是主要比较，冻结测试仅作报告。该设计只对齐下游任务样本曝光，不对齐预训练、token、优化器步数、FLOPs、实际时间、硬件、延迟、参数量或计算资源。

E. Statistical Analysis and Training Seeds / 统计分析与训练种子

Paired user-level bootstrap intervals [25] describe evaluation-sample uncertainty for the fixed seed42 Y96 and M1-Y96 models. Predictions share targets; users are resampled with replacement and all their examples enter both models' resamples. Each resample yields the binary M1-Y minus Y96 metric difference, whose 2.5th and 97.5th percentiles form the 95% interval. These intervals support only the Y-side binary comparison; cross-task-safe M1-N ranking results are reported without bootstrap intervals.

用户级配对 bootstrap 区间[25] 描述固定 seed42 Y96 与 M1-Y96 模型的评估样本不确定性。预测按相同目标对齐，以用户为单位有放回采样，并将入选用户的全部样本同时带入两个模型。每次重采样计算 M1-Y 减 Y96 的二分类指标差值，第 2.5 与 97.5 百分位构成 95% 区间。这些区间只支撑 Y 侧二分类比较；cross-task-safe M1-N 排序结果不报告 bootstrap 区间。

Zero-crossing Y-side intervals indicate uncertainty, not equivalence. Full trajectories use training seed42; seeds 43/44 independently replicate Y96, N96, and M1-96 at 96k on validation and frozen test under k5/k20/k50. Cross-task-safe

ranking summaries report point estimates, equalweight means, sample SD with $n=3$ and $ddof=1$, and per-seed directions. They describe training-run variability, not confidence intervals, significance, equivalence, or a multi-seed trajectory. Candidate-generation seed42 is recorded separately from the three training seeds.

Y 侧跨零区间表示不确定，并不证明等价。完整轨迹使用训练 seed42；seed43/44 在 96k 独立复现 Y96、N96 与 M1-96，并覆盖验证、冻结测试及 k5/k20/k50。cross-tasksafe 排序汇总报告点估计、 $n=3$ 且 $ddof=1$ 的等权均值与样本标准差，以及逐 seed 方向。它们描述训练运行间变异，不是置信区间、显著性判断、等价性判断或多 seed 轨迹。候选生成 seed42 与三个训练 seed 分别记录。

VI. Results / 实验结果

We first compare the capabilities measured under Y and N, then examine exposure response, specialist-multitask relationships, candidate protocols, and the N-SASRec comparison. MovieLens supplies the full seed42 trajectories and independent seed43/44 replications at 96k. Validation guided decisions; frozen-test results are report-only held-out evidence. Amazon provides an earlier-point ranking-side check.

结果先比较 Y 与 N 监督下的被测能力，再考察曝光响应、专家与多任务关系、候选协议及 N 与 SASRec 的比较。MovieLens 提供完整 seed42 轨迹和 seed43/44 在 96k 运行点的独立复现。验证集指导实验决策，冻结测试结果作为仅报告的留出证据。Amazon 提供较早运行点的排序侧检验。

A. Different Supervision and Observed Recommendation Capabilities / 不同监督与可观测推荐能力

Y and N show distinct measured capability profiles. Y96 reaches 0.7844 validation AUC on native MovieLens preference prediction. Through the P(Yes) bridge, its PopMatch-k5 validation HR@1 is 0.2211, compared with 0.6238 for N96 at the same nominal task-sample exposure. Native preference metrics, the Y-as-ranker trajectory, and validation/frozen-test Y96-N96 endpoints appear in Table IV.

Y 与 N 呈现不同的被测能力结构。Y96 在 MovieLens 原生偏好预测上的验证 AUC 为 0.7844；通过 P(Yes) 桥接后，其 PopMatch-k5 验证 HR@1 为 0.2211，而相同名义任务样本曝光下 N96 为 0.6238。原生偏好指标、Y-as-ranker 轨迹以及验证与冻结测试上的 Y96-N96 端点见 Table IV。

This contrast separates rating-derived preference discrimination from selecting the next observed event. Because target construction, task data, prompts, and scoring interfaces differ jointly, it compares complete supervision formulations rather than isolating one semantic cause.

TABLE IV MovieLens seed42 supervision profiles. Panel A reports Y-native preference metrics across validation exposure and the held-out Y96 endpoint. Panel B reports PopMatch-k5 Y-as-ranker validation exposure, the Y96/N96 validation contrast, and frozen-test endpoints. AUC/F1/accuracy and HR@1/NDCG@5/MRR belong to different interfaces and are not compared numerically across panels. / MovieLens seed42 监督能力结构。Panel A 报告 Y 原

生偏好指标的验证曝光轨迹及 Y96 留出测试端点；Panel B 报告 PopMatch-k5 下 Y-as-ranker 验证曝光轨迹、Y96/N96 验证对照及冻结测试端点。
AUC/F1/Accuracy 与 HR@1/NDCG@5/MRR 属于不同接口，不跨 panel 作数值比较。

Panel	Run	Exposure	Split	Interface	Metric 1	Value 1	Metric 2	Value 2	Metric 3	Value 3
A	Y24	24000	validation	Y-native preference	AUC	0.77613	F1	0.77917	Accuracy	0.71909
A	Y48	48000	validation	Y-native preference	AUC	0.78161	F1	0.78484	Accuracy	0.72304
A	Y96	96000	validation	Y-native preference	AUC	0.78435	F1	0.77832	Accuracy	0.72353
A	Y96	96000	test	Y-native preference	AUC	0.78535	F1	0.77802	Accuracy	0.72211
B	Y24	24000	validation	Y-as-ranker bridge	HR@1	0.21568	NDCG@5	0.60027	MRR	0.47041
B	Y48	48000	validation	Y-as-ranker bridge	HR@1	0.21656	NDCG@5	0.59946	MRR	0.46948
B	Y96	96000	validation	Y-as-ranker bridge	HR@1	0.22115	NDCG@5	0.60307	MRR	0.47418
B	N96	96000	validation	N-native ranking	HR@1	0.62379	NDCG@5	0.83029	MRR	0.77324
B	Y96	96000	test	Y-as-ranker bridge	HR@1	0.20652	NDCG@5	0.59216	MRR	0.46005
B	N96	96000	test	N-native ranking	HR@1	0.61004	NDCG@5	0.82190	MRR	0.76220

这一对照区分了评分产生的偏好判别与下一观测事件选择。由于目标构造、任务数据、提示和评分接口共同变化，比较针对完整监督形式，而非隔离某个语义因素的因果作用。

B. Task-Specific Response to Training Exposure / 不同任务对训练曝光的响应

Y-native validation AUC rises from 0.7761 at 24k to 0.7844 at 96k. F1 and the Y-as-ranker metrics do not improve uniformly, so the measured Y response is limited or uneven rather than demonstrably saturated.

Y 原生验证 AUC 从 24k 时的 0.7761 升至 96k 时的 0.7844。F1 与 Y-as-ranker 指标并未一致改善，因此 Y 的被测响应有限或不均匀，不能据此认定已经饱和。

N-native PopMatch-k5 validation HR@1 rises from 0.5774 at N24 to 0.6516 at N200, an absolute gain of 0.0742; the test trajectory has the same direction. Figure 1 shows the validation trend, while Table V retains the exact HR@1 values for both splits.

N 原生 PopMatch-k5 验证 HR@1 从 N24 的 0.5774 升至 N200 的 0.6516，绝对增量为 0.0742；测试轨迹方向相同。Figure 1 展示验证趋势，Table V 保留两个划分的精确 HR@1 数值。

Y and N therefore have different observed exposure-response profiles. This comparison concerns within-interface trajectories, not the numerical magnitudes of AUC and HR@1, and motivates matching corresponding task exposure in the specialist-multitask analysis.

Y 与 N 由此呈现不同的已观测曝光响应。这里比较的是各自接口内的轨迹，而非 AUC 与 HR@1 的数值大小，并据此在专家与多任务分析中对齐对应任务曝光。

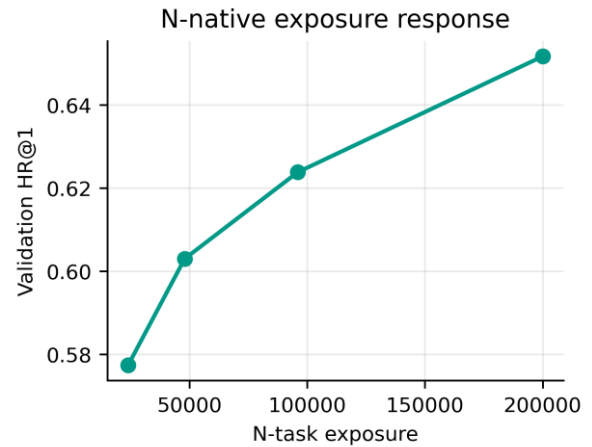


Fig. 1. MovieLens seed42 PopMatch-k5 N ranking performance across the evaluated task-exposure points. / MovieLens seed42 PopMatch-k5 下 N 排序表现随已测任务曝光点的变化。

C. Specialist and Multitask Models at Matched Per-Task Exposure / 每任务曝光对齐下的专家与多任务模型

At 96k expected exposure per task, M1 retains comparable measured Y-side preference capability across three training seeds. M1-Y validation AUC and F1 are slightly higher in each seed, whereas accuracy is lower by 0.00089 for seed44. Table VI reports the seed42 paired user bootstrap for this binary Y-side comparison: only the F1 interval is positive, while AUC and accuracy intervals cross zero. This supports capability preservation, not equivalence or positive transfer across training seeds.

在每任务预期曝光 96k 时，M1 在三个训练 seed 上保持了与 Y 专家接近的被测偏好能力。各 seed 的 M1-Y 验证 AUC 与 F1 略高，但 seed44 的 Accuracy 低 0.00089。Table VI 报告这一 Y 侧二分类比较的 seed42 用户级配对 bootstrap：仅 F1 区间为正，AUC 与 Accuracy 区间跨零。

这些证据支持能力保留，不证明等价或跨训练 seed 正迁移。

TABLE V MovieLens seed42 N/SASRec ranking on validation and test sets at approximately matched task-sample exposure (HR@1). / MovieLens seed42 验证集与测试集上，任务样本曝光近似对齐时的 N/SASRec 排序 (HR@1)。

N run	N exposure	SASRec exposure	Split	N HR@1	SASRec HR@1	Delta HR@1
N24	24000	24064	validation	0.57744	0.27313	0.30432
N24	24000	24064	test	0.56123	0.29040	0.27084
N48	48000	48128	validation	0.60300	0.29304	0.30996
N48	48000	48128	test	0.58696	0.30220	0.28476
N96	96000	96256	validation	0.62379	0.32811	0.29568
N96	96000	96256	test	0.61004	0.33093	0.27912
N200	200000	200000	validation	0.65163	0.47489	0.17674
N200	200000	200000	test	0.62819	0.45110	0.17709

TABLE VI MovieLens seed42 validation paired binary differences and user-bootstrap 95% intervals for Y96 and M1-Y96. Delta is M1-Y96 minus Y96. This table supports only the Y-side preference comparison. / MovieLens seed42 验证集上 Y96 与 M1-Y96 的配对二分类差值及用户级 bootstrap 95% 区间。差值为 M1-Y96 减 Y96。本表只支撑 Y 侧偏好比较。

Comparison	Metric	Delta	CI low	CI high
Y96 vs M1-96-Y	AUC	0.00248	-0.00167	0.00674
Y96 vs M1-96-Y	F1	0.00553	0.00096	0.01025
Y96 vs M1-96-Y	Accuracy	0.00460	-0.00089	0.01035

On the common cross-task-safe validation subset in Table VII, the seed42 standard-k5 N-specialist-M1 gap narrows from 48k to 96k per task across all three metrics. The HR@1 gap falls from +0.01072 to +0.00132; NDCG@5 and MRR show the same direction. Independent seeds 43 and 44 reproduce a small but consistently positive N advantage at 96k: Table VIII reports a three-seed validation HR@1 mean delta of +0.00708 with sample SD 0.00934 and 3/3 positive directions. The full exposure trajectory remains seed42-only.

在 Table VII 的共同 cross-task-safe 验证子集上，seed42 标准 k5 的 N 专家-M1 差距在三个指标上均从每任务 48k 到 96k 收窄。HR@1 差距由 +0.01072 降至 +0.00132，NDCG@5 与 MRR 方向相同。seed43 和 44 独立复现了 96k 时小幅但方向一致的 N 优势：Table VIII 报告三 seed 验证 HR@1 平均差值为 +0.00708、样本标准差为 0.00934，且 3/3 方向为正。完整曝光轨迹仍仅来自 seed42。

Frozen tests retain an N advantage in every seed at 96k. Table VIII reports a three-seed HR@1 mean delta of +0.01072 with sample SD 0.00271 and 3/3 positive directions. On the common seed42 test subset in Table VII, however, the HR@1 gap changes from +0.00939 at 48k to +0.01373 at 96k, and NDCG@5 and MRR likewise do not narrow. The validation-side exposure trajectory is therefore not reproduced on test.

96k 冻结测试在每个 seed 上均保留 N 优势。Table VIII 报告三 seed 的 HR@1 平均差值为 +0.01072、样本标准差为 0.00271，且 3/3 方向为正。然而，在 Table VII 的共同 seed42 测试子集上，HR@1 差距由 48k 的 +0.00939 变为

96k 的 +0.01373，NDCG@5 与 MRR 同样未收窄。因此，验证侧的曝光轨迹没有在测试集复现。

D. Robustness under Alternative Candidate Protocols / 不同候选协议下的鲁棒性

At 96k, k20 produces the largest N-specialist-M1 gap among the three evaluated protocols. The cross-task-safe three-seed summary in Table VIII gives HR@1 mean deltas of +0.08042 on validation and +0.07678 on frozen test. Across k5, k20, and k50, all 54 seed-level deltas are positive and all 18 protocol-split-metric summaries have 3/3 positive seed directions.

在 96k 时，k20 在三种已评估协议中产生最大的 N 专家-M1 差距。Table VIII 的 cross-task-safe 三 seed 汇总给出验证集 +0.08042、冻结测试 +0.07678 的 HR@1 平均差值。k5、k20 和 k50 合计 54 个 seed 级差值全部为正，18 个协议-划分-指标汇总项也全部呈现 3/3 seed 正向。

k50 also favors N across seeds, with three-seed HR@1 mean deltas of +0.00990 on validation and +0.01048 on test, but its relation to k5 is metric dependent. For seed43 test, k50 is smaller than k5 for HR@1 (+0.00560 versus +0.00849) and NDCG@5 (+0.00450 versus +0.00489), while its MRR delta is larger. k5 is popularity matched, whereas k20/k50 are random and all three protocols are separately constructed and non-nested. Candidate count, composition, sampling, and difficulty therefore vary jointly; the evidence establishes protocol dependence, not a causal or monotonic candidate-size effect.

TABLE VII Seed42 common cross-task-safe PopMatch-k5 comparison. Delta is N minus M1; change is delta96 minus delta48 within each split and metric. / seed42 共同 cross-task-safe PopMatch-k5 比较。差值为 N 减 M1；变化为同一划分和指标下 delta96 减 delta48。

Split	Exp. (k)	n	Metric	N	M1-N	Delta	Delta change
validation	48	5318	HR@1	0.60323	0.59252	0.01072	
validation	96	5318	HR@1	0.62223	0.62091	0.00132	-0.00940
validation	48	5318	NDCG@5	0.82000	0.81451	0.00549	
validation	96	5318	NDCG@5	0.82961	0.82784	0.00177	-0.00372
validation	48	5318	MRR	0.75953	0.75225	0.00728	
validation	96	5318	MRR	0.77233	0.77004	0.00230	-0.00498

test	48	5535	HR@1	0.58500	0.57561	0.00939	
test	96	5535	HR@1	0.60741	0.59368	0.01373	0.00434
test	48	5535	NDCG@5	0.80952	0.80499	0.00453	
test	96	5535	NDCG@5	0.82044	0.81439	0.00605	0.00151
test	48	5535	MRR	0.74575	0.73974	0.00601	
test	96	5535	MRR	0.76028	0.75221	0.00807	0.00206

TABLE VIII Cross-task-safe 96k N96-M1-N ranking deltas for training seeds 42/43/44. Mean and sample SD use $n=3$ and $ddof=1$; signs follow seed order. These are descriptive summaries. Protocols are separate and non-nested. / 训练 seed 42/43/44 的 cross-task-safe 96k N96-M1-N 排序差值。均值与样本标准差采用 $n=3$ 、 $ddof=1$ ，方向按 seed 顺序排列；这些量仅为描述性汇总。各协议分 别构造且非嵌套。

Split	Protocol	Metric	Delta	Mean	SD	Signs 42/43/44
validation	k5	HR@1	N96 - M1-N	0.00708	0.00934	+/+/+
validation	k5	NDCG@5	N96 - M1-N	0.00345	0.00288	+/+/+
validation	k5	MRR	N96 - M1-N	0.00461	0.00396	+/+/+
test	k5	HR@1	N96 - M1-N	0.01072	0.00271	+/+/+
test	k5	NDCG@5	N96 - M1-N	0.00528	0.00066	+/+/+
test	k5	MRR	N96 - M1-N	0.00700	0.00093	+/+/+
validation	k20	HR@1	N96 - M1-N	0.08042	0.02971	+/+/+
validation	k20	NDCG@5	N96 - M1-N	0.11826	0.01135	+/+/+
validation	k20	MRR	N96 - M1-N	0.08724	0.01896	+/+/+
test	k20	HR@1	N96 - M1-N	0.07678	0.03492	+/+/+
test	k20	NDCG@5	N96 - M1-N	0.11573	0.01356	+/+/+
test	k20	MRR	N96 - M1-N	0.08441	0.02143	+/+/+
validation	k50	HR@1	N96 - M1-N	0.00990	0.00696	+/+/+
validation	k50	NDCG@5	N96 - M1-N	0.01457	0.01310	+/+/+
validation	k50	MRR	N96 - M1-N	0.01830	0.00731	+/+/+
test	k50	HR@1	N96 - M1-N	0.01048	0.00444	+/+/+
test	k50	NDCG@5	N96 - M1-N	0.01418	0.01223	+/+/+
test	k50	MRR	N96 - M1-N	0.01806	0.00597	+/+/+

k50 在各 seed 上也偏向 N，三 seed HR@1 平均差值在验证与测试上分别为+0.00990 和+0.01048，但它与 k5 的相对关系依指标而变。seed43 测试中，k50 的 HR@1 差值（+0.00560）和 NDCG@5 差值（+0.00450）小于 k5（+0.00849 和+0.00489），MRR 差值则更大。k5 采用流行度匹配，k20/k50 采用随机候选，三种协议分别构造且非嵌套。候选数量、组成、采样和难度共同变化，因此证据只支持协议依赖性，不支持候选规模的因果或单调效应。

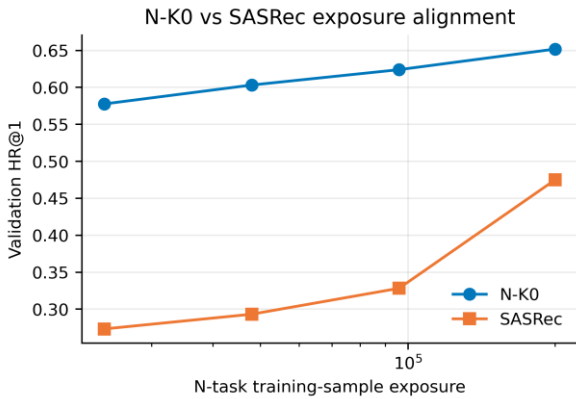


Fig. 2. MovieLens seed42 PopMatch-k5 validation HR@1 at approximately matched task-sample exposure. The N-K0 legend denotes the N24/N48/N96/N200 series. / MovieLens seed42 PopMatch-k5 验证 HR@1，按任务样本曝光近似对齐。图例 N-K0 指 N24/N48/N96/N200 系列。

E. Exposure-Aware Comparison with SASRec / 曝光感知的 SASRec 比较

N-trained LlamaRec leads SASRec at all four approximately matched seed42 downstream task-sample exposure points. On validation, the HR@1 gap narrows from +0.2957 at 96k to +0.1767 at 200k. Table V retains exact exposures and validation/frozen-test values; Figure 2 shows the validation trajectories.

接受 N 任务训练的 LlamaRec 在四个 seed42 下游任务样本曝光近似匹配点均领先 SASRec。验证 HR@1 差距从 96k 时的+0.2957 缩小到 200k 时的+0.1767。Table V 保留精确曝光以及验证与冻结测试数值，Figure 2 展示验证轨迹。

SASRec nevertheless improves strongly with additional supervision. The comparison therefore pairs N’s lead with the baseline’s continuing response and aligns only downstream task-sample exposure, not pretraining, tokens, updates, FLOPs, time, or end-to-end resource efficiency.

SASRec 仍随额外监督明显改善。因此，这一比较同时保留 N 的领先与基线的持续响应，并且只对齐下游任务样本曝光，不对齐预训练、token、更新次数、FLOPs、时间或端到端资源效率。

TABLE IX Amazon Musical Instruments seed42 full-test PopMatch-k5 ranking for the evaluated earlier operating points. / Amazon Musical Instruments seed42 在已评估较早运行点上，任务特定或满足跨任务安全条件模型的完整测试集 PopMatch-k5 排序。

Run	Samples	HR@1	NDCG@5	MRR
base	57439	0.35730	0.68295	0.57896
y	57439	0.22979	0.60997	0.48303

n	57439	0.46688	0.74201	0.65698
sasrec	57439	0.17566	0.56851	0.42952

F. Cross-Dataset Directional Validation / 跨数据集方向性检验

At the evaluated earlier seed42 operating points on Amazon Musical Instruments, N reaches 0.4669 HR@1 on 57,439 PopMatch-k5 test examples, compared with 0.3573 for Base, 0.2298 for Y-as-ranker, and 0.1757 for SASRec. Table IX reports these four cross-task-safe or task-specific model results with HR@1, NDCG@5, and MRR.

在 Amazon Musical Instruments 已评估的较早 seed42 运行点上, N 在 57,439 个 PopMatch-k5 测试样本中达到 0.4669 HR@1, Base、Y-as-ranker 与 SASRec 分别为 0.3573、0.2298 和 0.1757。Table IX 报告这四个满足跨任务安全或任务特定条件的模型结果, 包括 HR@1、NDCG@5 与 MRR。

Amazon therefore provides ranking-side directional support at available earlier points, not a second exposure curve, three-seed 96k replication, hard-candidate study, or native-preference validation.

因此, Amazon 仅在现有较早运行点提供排序侧方向性支撑, 不构成第二条曝光曲线、三 seed 的 96k 复现、较难候选研究或原生偏好验证。

VII. Discussion / 讨论

The findings tie three parts of a recommendation comparison to one another: supervision formulation defines the prediction question, task exposure locates the adaptation point, and candidate protocol defines the measured ranking conditions. Their implications differ from the numerical results and are organized below by the study's contribution hierarchy.

这些发现将推荐比较中的三个方面联系起来: 监督形式定义预测问题, 任务曝光确定适配运行点, 候选协议界定排序测量条件。下文按研究贡献层级讨论其含义, 而不重复结果数值。

A. Supervision Formulation and Capability Interpretation / 监督形式与能力解释

Recommendation capability is meaningful only relative to a prediction question. Y estimates rating-derived preference, whereas N selects the next observed interaction. Native preference discrimination and bridge ranking therefore assess different behavior: a model can be useful for preference while being less suited to next-event selection.

推荐能力只有相对于具体预测问题才有明确含义。Y 估计评分产生的偏好, N 选择下一观测交互。原生偏好判别与桥接排序因此评估不同的行为: 模型可以适合偏好判断, 却不一定适合下一事件选择。

A benchmark's target, eligible events, prompt, and scoring rule jointly determine the capability reflected by its score. Reporting native and bridge results exposes distinctions that one ranking number can hide and keeps the Y/N contrast attached to complete formulations rather than a single semantic cause.

基准的目标、合法事件、提示和评分规则共同决定分数反映的能力。并列报告原生与桥接结果, 可以呈现单排序数值容易掩盖的区别, 并将 Y/N 对照限定为完整形式比较, 而非单一语义原因。

B. Exposure and Specialist-Shared Modeling / 曝光与专家共享建模

Exposure locates a comparison along the observed adaptation trajectory. Seed42 shows continued N improvement but limited or uneven Y gains, so an early checkpoint need not describe the later model relationship. Treating task exposure as an experimental variable prevents an operating point from being mistaken for a stable property of the task.

曝光确定比较位于已观测适配轨迹的哪个位置。seed42 显示 N 持续改善, 而 Y 增益有限或不均匀, 因此早期检查点不必代表后续模型关系。把任务曝光作为实验变量, 可以避免将某个运行点误解为任务的稳定属性。

Task-specific accounting distinguishes M1's total exposure from the supervision allocated to each interface and makes SASRec comparisons interpretable across batch sizes. This axis is cumulative downstream consumption, including repetitions, rather than unique data or computation. Stating it beside performance makes the budget condition of each advantage visible.

按任务统计可以区分 M1 总曝光与分配给各接口的监督, 也使不同 batch size 下的 SASRec 比较具有明确口径。这一轴表示包含重复样本的累计下游消费, 而非唯一数据量或计算量。将其与表现并列报告, 可明确每项优势的预算条件。

The standard-k5 validation trajectory and independent 96k replications together show that specialist-shared relations are conditional on adaptation point. M1 retains comparable Y-side capability at 96k, while frozen tests still favor N. A shared adapter should therefore be assessed by capability retained at a stated exposure and protocol, not by an unconditional success/failure label. The full trajectory remains seed42-only.

标准 k5 验证轨迹与 96k 独立复现共同表明, 专家共享关系受适配运行点限定。M1 在 96k 保留接近的 Y 侧能力, 冻结测试仍偏向 N。因此, 共享 adapter 应按明确曝光和协议下保留的能力评估, 而不宜作无条件成败判断; 完整轨迹仍仅来自 seed42。

C. Candidate Protocol and Comparison Scope / 候选协议与比较范围

At 96k, k20 exposes a larger N advantage than standard k5, whereas k50 is smaller than k20 and not uniformly larger than k5. A standard-k5 result alone therefore cannot characterize specialist performance retained under other candidate conditions. The relation is protocol-conditioned, not monotonic in candidate count.

在 96k 时, k20 呈现比标准 k5 更大的 N 优势, k50 则小于 k20 且并非总大于 k5。因此, 仅凭标准 k5 结果不能刻画其他候选条件下保留的专家表现; 该关系受协议限定, 不随候选数量单调变化。

Fixed candidates within a protocol support paired model comparison, while cross-protocol differences describe sensitivity to evaluated conditions. Because the sets are independently constructed and non-nested, count, composition, sampling, and difficulty change jointly. This separates within-protocol evidence from unsupported candidate-count causality.

协议内部固定候选支持模型配对比较, 跨协议差异则描述对评估条件的敏感性。由于候选集独立构造且非嵌套, 数量、组成、采样和难度共同变化。这一区分保留协议内证据, 并排除未经支持的候选数量因果解释。

D. Exposure-Aligned Cross-Model Baseline Comparison / 曝光对齐下的跨模型基线比较

Across four approximately matched seed42 points, N leads the evaluated SASRec baseline, while SASRec improves substantially and narrows the largest-exposure gap. The former describes performance under limited downstream task supervision; the latter describes the baseline's response to receiving more supervision. Both are needed to interpret the cross-model comparison.

在四个 seed42 曝光近似匹配点上, N 领先被评估的 SASRec 基线, 同时 SASRec 明显改善并缩小最大曝光点的差距。前者描述有限下游任务监督下的表现, 后者描述基线获得更多监督后的响应; 两者缺一不可。

Task-sample exposure is a useful but partial cross-model axis. It describes adaptation from the LLM's pretrained starting point and does not match pretraining, tokens, updates, FLOPs, time, serving, or deployment cost. The evidence supports an exposure-aligned performance comparison, not general resource efficiency.

任务样本曝光是有用但局部的跨模型比较轴。它描述 LLM 从预训练起点进行的适配, 并不匹配预训练、token、更新、FLOPs、时间、服务或部署成本。现有证据支持曝光对齐的表现比较, 而非一般资源效率结论。

MovieLens supplies the detailed within-domain exposure and protocol analysis, whereas Amazon checks ranking directions only at available earlier seed42 points. The two domains therefore provide complementary detailed and directional evidence; untested cross-domain trajectories remain outside the present claims.

MovieLens 提供详细的领域内曝光与协议分析, Amazon 仅在现有较早 seed42 运行点检验排序方向。两个领域由此分别提供详细证据与方向证据, 未经检验的跨域轨迹不属于本文结论。

Recommendation capability and relative performance should thus be interpreted within the evaluated formulation, cumulative per-task exposure, and candidate protocol. The cross-model trajectory adds that an observed N advantage can coexist with continued improvement of the sequential baseline.

因此, 推荐能力与相对表现应结合已评估的监督形式、累计每任务曝光和候选协议解释。跨模型轨迹还表明, 已观测的 N 优势可以与序列基线的持续改善同时存在。

VIII. Limitations / 局限性

Only the 96k operating point has independent training seeds 42/43/44; full MovieLens trajectories remain seed42-only. Cross-task-safe ranking means and sample SD describe $n=3$ training-run variability and have no bootstrap intervals, whereas the retained seed42 paired bootstrap describes evaluation-sample uncertainty only for the fixed Y-side models. Neither establishes cross-seed significance. Validation guided selection and frozen test provides held-out reporting. No equivalence margin or allcomparison multiplicity adjustment was used. Behavior beyond Y96/N200 is unknown, and exposure changes coverage and optimization history together. M1 per-task counts are schedule expectations conditional on correct resume skipping, without a retained per-example trace.

只有 96k 运行点包含训练 seed42/43/44 的独立复现, 完整 MovieLens 轨迹仍仅来自 seed42。cross-task-safe 排序均值与样本标准差描述 $n=3$ 训练运行间变异且没有 bootstrap 区间, 保留的 seed42 配对 bootstrap 只描述固定 Y 侧模型的评估样本不确定性; 二者均不提供跨 seed 显著性结论。验证集指导选择, 冻结测试用于留出报告。研究未设等价界限, 也未对全部比较作多重性校正。Y96/N200 之外的行为未知, 曝光还同时改变覆盖和优化历程。M1 每任务计数为以正确续训跳过为条件的调度预期, 缺少逐样本轨迹。

Shared-model ranking comparisons use high-coverage cross-task-safe subsets rather than the entire N holdouts because Y and N use different task-specific temporal splits. This evaluation-side restriction ensures that an N target is not exposed through the shared adapter's Y-training stream; M1 was not trained under a joint temporal cutoff. Retained coverage ranges from 93.71% to 98.68%, so the conclusions apply to these retained populations. PopMatch-k5 and random k20/k50 are non-nested: count, composition, sampling, and difficulty vary jointly. PopMatch uses full-corpus popularity retrospectively, and sampled ranking is not exhaustive full-catalog evaluation.

由于 Y 与 N 使用不同的任务特定时间划分, 共享模型的排序比较采用高覆盖率 cross-task-safe 子集, 而不是完整 N 留出集。这一评估侧限制确保 N 目标不会经共享 adapter 的 Y 训练流被接触; M1 并未在联合时间截断下训练。保留覆盖率为 93.71% 至 98.68%, 因此相关结论适用于这些保留人群。PopMatch-k5 与随机 k20/k50 非嵌套, 候选数量、组成、采样和难度共同变化。PopMatch 回顾性使用全语料流行度, 采样候选排序也并非穷尽全目录评估。

Amazon support is limited to seed42 ranking directions at earlier test operating points. It does not replicate exposure trajectories, the three-seed 96k point, hard-candidate robustness, or native preference validation. Detailed exposure and protocol conclusions therefore remain specific to MovieLens.

Amazon 支撑限于较早测试运行点的 seed42 排序方向, 未复现曝光轨迹、三 seed 的 96k 运行点、较难候选稳健

性或原生偏好验证。因此，详细曝光与协议结论仍限定于 MovieLens。

Task-sample matching does not remove resource differences. The LLM starts pretrained; SASRec learns sequence representations from recommendation data. Exposure matches neither pretraining nor tokens, updates, FLOPs, wall-clock time, serving latency, or deployment cost. Claims concern ranking under approximately matched downstream task-sample exposure in the evaluated implementations and regimes.

任务样本匹配不消除资源差异。LLM 从预训练底座开始，SASRec 从推荐数据学习序列表示。曝光不对齐预训练、token、更新、FLOPs、实际时间、服务延迟或部署成本。本文结论仅涉及被评估实现和训练条件下，下游任务样本曝光近似匹配时的排序表现。

IX. Conclusion / 结论

This study examined LLM recommendation conclusions across complete supervision formulations, cumulative downstream task-sample exposure, and candidate protocols. Within one adaptation setting, rating-derived preference (Y), next-interaction prediction (N), and a shared adapter (M1) represent different prediction capabilities rather than a causally isolated semantic contrast.

本研究考察完整监督形式、累计下游任务样本曝光和候选协议下的 LLM 推荐结论。在共同适配设置中，评分产生的偏好 Y、下一交互预测 N 和共享 adapter M1 代表不同预测能力，而非某个被因果隔离的语义对照。

On the common cross-task-safe subset, the seed42 specialist-M1 validation gap narrows from 48k to 96k across all three ranking metrics, while the test subset does not reproduce that narrowing. Independent seeds replicate only the 96k operating point, where ranking consistently favors N and M1 preserves comparable Y-side capability. Thus, specialist-shared performance is exposure and protocol conditioned rather than universally equivalent. Across four approximately matched downstream exposure points, N-trained LlamaRec leads SASRec, but SASRec continues improving and narrows the later gap; this is a task-sample exposure comparison, not a resource-efficiency claim.

在共同 cross-task-safe 子集上，seed42 专家-M1 验证差距在三个排序指标上均从 48k 到 96k 收窄，而测试子集没有复现这一收窄。独立训练 seed 只复现 96k 运行点；此时排序一致偏向 N，M1 则保持接近的 Y 侧能力。因此，专家-共享表现受曝光和协议限定，并非普遍等价。在四个下游曝光近似匹配点上，N 训练的 LlamaRec 领先 SASRec，但 SASRec 继续改善并缩小后期差距；这属于任务样本曝光比较，不是资源效率结论。

Claims about adapted language models for recommendation should therefore report the prediction formulation, cumulative downstream task-sample exposure, and candidate evaluation protocol together. These conditions define which capability and relative model performance the evidence supports.

因此，关于推荐适配语言模型的结论应同时报告预测形式、累计下游任务样本曝光和候选评估协议。这些条件共同界定现有证据支持何种能力与模型相对表现。

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